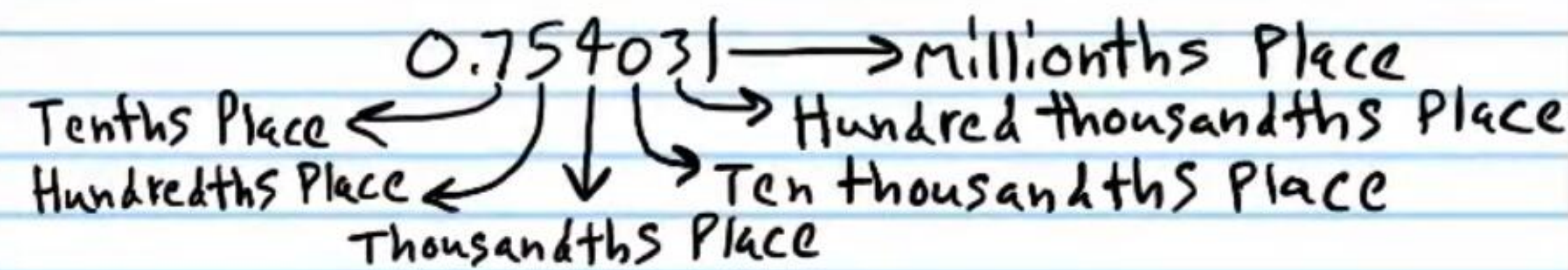


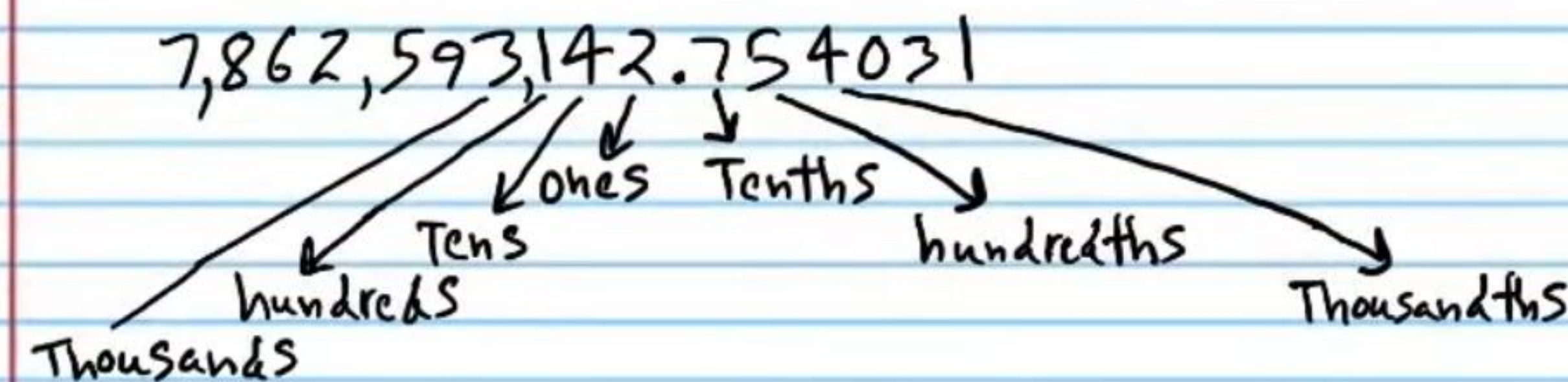
Class 18: An Introduction to Decimals

Example:

$$0.754031 = \frac{7}{10} + \frac{5}{100} + \frac{4}{1,000} + \frac{0}{10,000} + \frac{3}{100,000} + \frac{1}{1,000,000}$$



$$0.754031 = \frac{7}{10} + \frac{5}{10^2} + \frac{4}{10^3} + \frac{0}{10^4} + \frac{3}{10^5} + \frac{1}{10^6}$$



Convert the following decimals to fraction form.

① 0.7

④ 0.3

② 0.02

⑤ 0.005

③ 0.0008

⑥ 0.000004

⑦ 0.851

⑪ 0.00513

⑧ 0.7132

⑫ 0.00063

⑨ 0.21

⑬ 0.3000

⑩ 0.351888

⑭ 0.56080

⑪ 0.98903

⑮ 0.00700

⑫ 0.003506

⑯ 0.080400

Convert the following fractions to decimal form.

⑰ $\frac{7}{10}$

⑬ 0.091

⑱ $\frac{7}{100}$

⑭ 0.000151

⑲ $\frac{89}{100,000}$

⑮ 0.00007009

⑳ $\frac{756}{1,000,000}$

⑯ 0.0705

㉑ $\frac{894}{1000}$

$$\textcircled{28} \frac{9}{10,000}$$

$$\textcircled{33} \frac{708050}{1,000,000}$$

$$\textcircled{29} \frac{9}{10}$$

$$\textcircled{34} \frac{2}{5}$$

$$\textcircled{30} \frac{73}{10,000}$$

$$\textcircled{35} \frac{11}{25}$$

$$\textcircled{31} \frac{229}{10,000,000}$$

$$\textcircled{36} \frac{5}{8}$$

$$\textcircled{32} \frac{73,129}{100,000}$$

Convert the following fractions to decimal form by using the long division algorithm.

$$\textcircled{37} \frac{5}{8}$$

$$\textcircled{38} \frac{4}{5}$$

$$\textcircled{39} \frac{3}{4}$$

$$\textcircled{40} \frac{1}{2}$$

$$\textcircled{41} \frac{3}{8}$$

$$\textcircled{42} \frac{1}{4}$$

$$\textcircled{43} \frac{3}{16}$$

$$\textcircled{44} \frac{7}{20}$$

$$\textcircled{45} \frac{2}{3}$$

$$\textcircled{46} \frac{5}{9}$$

$$(47) \frac{8}{9}$$

$$(48) \frac{1}{3}$$

$$(49) \frac{1}{15}$$

$$(50) \frac{1}{18}$$

$$(51) \frac{5}{6}$$

$$(52) \frac{11}{15}$$

$$(53) \frac{1}{6}$$

$$(54) \frac{5}{12}$$

$$(55) \frac{2}{11}$$

$$(56) \frac{19}{22}$$

$$(57) \frac{10}{11}$$

$$(58) \frac{7}{22}$$

$$\textcircled{59} \frac{1}{11}$$

$$\textcircled{60} \frac{1}{12}$$

$$\textcircled{61} \frac{6}{7}$$

$$\textcircled{62} \frac{3}{7}$$

$$\textcircled{63} \frac{4}{7}$$

$$\textcircled{64} \frac{1}{7}$$

$$\textcircled{65} \frac{8}{13}$$

$$\textcircled{66} \frac{5}{14}$$

Convert the following decimals to Mixed Numbers

(67) 8.124

(68) 56.1

(69) 382.0051

(70) 23.0000076

Convert the following Mixed Numbers to Decimals

(71) $3\frac{7}{10}$

(72) $6\frac{3}{8}$

(73) $2\frac{1}{5}$

(74) $23\frac{1}{6}$

(75) $718\frac{4}{9}$

(76) $81\frac{2}{7}$